Project Exercises

Due Date: April 29, 2026 (No Penalty Extension to May 4)

Purpose

This assignment asks you to carry out the full data pipeline for an empirical research project: collect raw data from a primary source, clean and reshape it, construct analysis variables, run a validation regression, and produce a publication-quality figure. This is the daily work of empirical research—and the part that takes the most time.

You will also practice the professional workflow skills that make research reproducible: organized directory structures, version control with Git and GitHub, and compiled documents in LaTeX.

Choose Your Path

Path 1: Your Own Project. Use the data and research question from your project proposal. Demonstrate the collect → clean → visualize pipeline with your own data. You must download or access data from a primary source, clean and construct at least one analysis variable. Your pipeline must incorporate at least two of the following three data wrangling operations: crosswalking/merging, aggregation/collapsing, and reshaping. Produce at least one figure and one regression table that are automated/draft ready. Your write-up should describe what you did at each stage and what you learned about your data.

Path 2: Guided Exercise. Construct an immigration shift-share (Bartik) instrumental variable using U.S. Census microdata from IPUMS, following the approach of Kim, Merritt, and Peri (NBER Working Paper 32526). Details on the next page.

Both paths are graded on the same rubric. Path 2 provides more structure, but may be more complex; Path 1 requires more initiative.

Grading: The Project Proposal and Project Exercises together are worth 30% of your final grade, split **2:1 in your favor**—whichever you score higher on counts for 20%, the other for 10%.

Requirements (Both Paths)

1. **Directory structure.** Organize your project into a sensible folder structure. A suggested layout:

```
project/
  data/raw/          <- original downloaded data (never modify)
  data/clean/        <- processed datasets
  code/              <- your .do, .py, or .R files
  output/figures/    <- figures
  output/tables/     <- tables
  docs/              <- your write-up (.tex and .pdf)
  README.md          <- brief description of the project
```

You may adjust this structure, but your files must be logically organized—not dumped in one folder.

2. **GitHub.** Your project must live in a GitHub repository with a meaningful commit history. I should see multiple commits reflecting your progress—not a single commit at the end. Include a `README.md` that describes the project and explains how to reproduce your results.

3. **LaTeX.** Your write-up must be compiled in LaTeX (or R Markdown/Quarto with PDF output). Submit both the source file and the compiled PDF.

4. **Reproducibility.** Your code must run from start to finish without errors, given the raw data and any provided crosswalk files. Submit a log file (Stata) or console output documenting the complete run.

What to Submit (on Canvas)

- A link to your GitHub repository
- Your compiled PDF write-up (1–3 pages, excluding figures and tables)
- Your code file(s) and log file
- At least one figure and one regression table

Collaboration & AI Policy

Same policy as problem sets. You may use AI tools for debugging and code assistance, but you must document your use. Include an AI log or attest to non-use in your write-up.

Guided Exercise: Constructing an Immigration Shift-Share IV

Your Goal

Construct a shift-share (Bartik) instrumental variable for immigration across U.S. commuting zones, using five decades of Census microdata (1970–2010). Then use the instrument to estimate the causal effect of immigration on local wages.

Background

Immigrants from a given origin country tend to settle where earlier immigrants from the same country already live—driven by established ethnic networks, not only current local labor demand. A shift-share instrument exploits this pattern. The instrument for commuting zone c in decade t is:

$$\widetilde{B}_{c,t} = \frac{1}{\text{pop}_{c,1970}} \sum_{o} \underbrace{\frac{\text{imm}_{c,o,1970}}{\text{imm}^{\text{national}}_{o,1970}}}_{\text{1970 share}} \times \underbrace{\left(\text{imm}^{\text{national}}_{o,t} - \text{imm}^{\text{national}}_{o,t-1}\right)}_{\text{national change, decade } t}$$

A detailed description of the methodology, notation, and identification logic is provided in Appendix A.

Reference: Kim, Gueyon, Cassandra Merritt, and Giovanni Peri. "Measuring and Predicting 'New Work' in the United States: The Role of Local Factors and Global Shocks." NBER Working Paper No. 32526, May 2024.

Provided Files

File	Description
`cw_bpld_kmp.dta`	Birthplace crosswalk: `BPLD` → ∼84 origin groups (`kmp_ctry`). Code 78 = US-born, 82 = other (drop).

Students collect: 5 IPUMS Census extracts (1970–2010), 4 Dorn geography crosswalks from `https://www.ddorn.net/data.htm`.

Pipeline Overview

The exercise has six steps. Steps 1–5 construct the dataset; Step 6 produces results. Each step saves a named output file, so you can work incrementally and verify as you go (see Appendix B for checkpoint statistics). The key Stata operations at each step are noted in parentheses.

1. **Process each decade → CZ × origin intermediates.** Load each IPUMS extract, apply sample restrictions (`drop if`), convert geography to commuting zones (`joinby`), classify by origin (`merge m:1`), construct wages, and collapse to CZ × origin (`collapse`). Repeat for all five decades, varying only the geography step. → `cz_o_YEAR.dta` *p. 4*

2. **Compute national immigration flows (shifts).** Append the five intermediate files (`append`), collapse to national origin-level totals, reshape wide (`reshape wide`), and compute decade-to-decade changes. → `origin_shifts.dta` *p. 5*

3. **Compute CZ-level actual statistics.** Append the same intermediate files, collapse across origins to CZ × year totals. → `cz_actual_panel.dta` *p. 5*

4. **Construct the shift-share instrument.** Load the 1970 intermediate file, merge national shifts (`merge m:1`), compute shares and predicted inflows (`gen`), sum across origins (`collapse`), normalize, and reshape to panel (`reshape long`). → `cz_predicted_panel.dta` *p. 6*

5. **Assemble the analytic dataset.** Merge actuals and predicted (`merge 1:1`), compute immigration and wage changes. → `cz_analytic_panel.dta` *p. 6*

6. **Estimation and visualization.** Summary table (`tabstat`), map (`spmap` or Python), scatterplots (`twoway scatter`), first-stage and IV regressions (`ivreg2`), interpretation. *p. 7*

The detailed methodology behind the shift-share formula is in Appendix A. Checkpoint statistics for verifying your intermediate files are in Appendix B.

The Pipeline: Detailed Steps

Step 1: Process Each Decade → CZ × Origin

Goal: For each Census decade, produce one intermediate dataset at the commuting zone × origin level, containing population counts and average wages.

Operations (repeat for each decade):

1. Load IPUMS extract.

2. Restrict sample: drop AK/HI (`statefip` 2, 15), group quarters (`GQ` 0, 3, 4), age < 16.

3. Convert geography to commuting zones: construct the merge key, `joinby` the Dorn crosswalk, create adjusted weights (`aperwt` = allocation factor × `perwt`). Note: `joinby` is a many-to-many merge—observation counts will increase because some geographic units span multiple commuting zones.

4. Classify by origin: merge `m:1` the birthplace crosswalk on `bpld`, keeping only matched observations (`keep(3)`). Drop `kmp_ctry == 82` (other/unspecified origins).

5. Construct hourly wages: convert intervalled weeks (`wkswork2`) and hours (`hrswork2` or `uhrswork`) to continuous measures using midpoints. Compute `wage` = `incwage`/(`hours` × `weeks`). Drop top-coded, zero, and extreme values.

6. Collapse to CZ × origin level: sum population, weighted mean wage.

   Hint: To get both a raw population sum and a population-weighted mean wage in a single `collapse`, use `(rawsum)` for the population variable and `(mean)` with `[pw=aperwt]` for wages. The `rawsum` option computes an unweighted sum even when probability weights are specified.

7. Save as `cz_o_YEAR.dta`.

What varies by decade: Geography variable and crosswalk file, merge key formula, allocation factor name (`afact` in 1970, `afactor` in 1980+), hours variable (intervalled in 1970, continuous in 1980+).

Decade	Geography var	Merge key	Crosswalk	Alloc. factor
1970	`CNTYGP97`	rename to `cty_grp70`	`cw_ctygrp1970_czone_corr.dta`	`afact`
1980	`CNTYGP98`	$1000 \times$ `statefip` + `cntygp98`	`cw_ctygrp1980_czone_corr.dta`	`afactor`
1990	`PUMA`	$10000 \times$ `statefip` + `puma`	`cw_puma1990_czone.dta`	`afactor`
2000	`PUMA`	$10000 \times$ `statefip` + `puma`	`cw_puma2000_czone.dta`	`afactor`
2010	`PUMA`	$10000 \times$ `statefip` + `puma`	`cw_puma2000_czone.dta` (same)	`afactor`

Hint: This step is repetitive by design. Write a loop with if-blocks for the decade-specific geography, rather than copy-pasting five blocks of code.

Step 2: Compute National Immigration Flows (Shifts)

Goal: From the CZ $\times$ origin intermediate files, compute national-level immigrant totals by origin for each decade, then compute decade-to-decade changes.

Operations:

1. Append all five intermediate files (`append using`).
2. Drop US-born (`drop if kmp_ctry == 78`).
3. Collapse to origin $\times$ year level (`collapse (sum) ..., by(kmp_ctry year)`): sum population across all CZs to get national totals per origin.
4. Reshape wide (`reshape wide natl_imm, i(kmp_ctry) j(year)`): one row per origin, columns for each decade's national total.
5. Compute shifts (`gen`): $\Delta M_{o,t} = \texttt{natl_imm}_{o,t} - \texttt{natl_imm}_{o,t-1}$.
6. Save as `origin_shifts.dta`.

Step 3: Compute CZ-Level Actual Statistics

Goal: From the same intermediate files, compute CZ-level totals: population, immigrant count, and average wage per CZ $\times$ year.

Operations:

1. Append all five `cz_o_YEAR` files.
2. Flag immigrants (`kmp_ctry` $\neq 78$).
3. Collapse to CZ $\times$ year level: sum total population and immigrant population, weighted mean wage.

   Hint: Same `(rawsum)` + `(mean) [pw=]` pattern as Step 1, but now the weight variable is your population count (`pop`), not the original `aperwt` (which no longer exists after the Step 1 collapse).

4. Save as `cz_actual_panel.dta`.

Step 4: Construct the Shift-Share Instrument

Goal: Using 1970 base-year shares and national shifts, compute predicted immigration inflows per CZ per decade.

Operations:

1. Load the 1970 CZ × origin file (`cz_o_1970.dta`).
2. Compute 1970 CZ total population (`bysort czone: egen pop1970 = total(pop)`)—sum across all origins including US-born.
3. Keep only immigrants (`drop if kmp_ctry == 78`).
4. Merge in national shifts (`merge m:1 kmp_ctry using origin_shifts.dta`).
5. Compute shares (`gen`): $s_{c,o} = \texttt{pop}_{c,o} \,/\, \texttt{natl_imm}_{o,1970}$.
6. Compute predicted inflows for each decade (`gen`): $\texttt{predicted}_{c,o,t} = s_{c,o} \times \texttt{shift}_{o,t}$.
7. Collapse to CZ level: sum predicted inflows across origins, keeping `pop1970` (`collapse (sum) predicted* (first) pop1970, by(czone)`).
8. Normalize by 1970 CZ population (`replace`): $\widetilde{B}_{c,t} = \texttt{predicted}_{c,t} \,/\, \texttt{pop}_{c,1970}$.
9. Reshape to long (`reshape long predicted, i(czone pop1970) j(year)`). Include `pop1970` in the `i()` stub so it survives the reshape.
10. Rename to match the analysis variables: `rename predicted imm_shift`.
11. Save as `cz_predicted_panel.dta`.

Step 5: Assemble the Analytic Dataset

Goal: Merge actual CZ statistics with the shift-share instrument. Compute changes.

Operations:

1. Load `cz_actual_panel.dta` (which includes all five decades, including 1970).
2. Merge (`merge 1:1 czone year using cz_predicted_panel.dta`). The 1970 observations will be unmatched (no instrument for the base year)—this is expected.
3. Compute lagged values. Sort by CZ and year (`sort czone year`), then use `bysort czone (year)` to generate lagged immigration and wages from the prior decade (e.g., `gen imm_lag = imm[_n-1]`). Because 1970 is still in the data, the 1980 observations will correctly pick up 1970 as their lag.

4. Compute actual immigration change (`gen imm_actch`):

$$\Delta \text{imm}_{c,t} = \frac{\text{imm}_{c,t} - \text{imm}_{c,t-1}}{\text{pop}_{c,1970}}$$

5. Compute wage change (`gen wage_ch`): $\Delta \text{wage}_{c,t} = \text{wage}_{c,t} - \text{wage}_{c,t-1}$.

6. Drop 1970 (it has no instrument and no lagged change—it served only as the base for lags). Drop the lag variables.

7. Save as `cz_analytic_panel.dta`.

Step 6: Estimation and Visualization

Deliverables:

1. **Summary table** (`tabstat ..., by(year)`): Mean, SD, min, max of the shift-share instrument, actual immigration change, and wage change—by decade.

2. **Map** (`spmap` in Stata or `geopandas` in Python): Choropleth of the shift-share instrument across commuting zones for at least one decade. You will need a commuting zone shapefile (available from Dorn's data page).

3. **First-stage scatterplots** (`twoway scatter ..., by(year)`): Shift-share instrument vs. actual immigration change, paneled by decade. Add fitted lines (`lfit`).

4. **First-stage regression** (`reg` or `ivreg2`):

$$\Delta \text{imm}_{c,t} = \beta \, \widetilde{B}_{c,t} + \gamma_t + \varepsilon_{c,t}$$

   Include year fixed effects (`i.year`) and cluster standard errors at the CZ level (`cluster(czone)`). Then add CZ fixed effects.

5. **Wage regressions** (`ivreg2` or `ivregress 2sls`): OLS and IV/2SLS of $\Delta \text{wage}_{c,t}$ on $\Delta \text{imm}_{c,t}$, instrumenting with $\widetilde{B}_{c,t}$. Year FE, clustered SEs. Report both in a table (`esttab` or `estout`).

6. **Interpret:** Is the instrument strong? (Check the first-stage F-statistic.) How do OLS and IV compare? What does the difference suggest about the direction of bias?

Extensions (for Exceptional credit)

- **Coefficient plot** of regression results across specifications or decades.
- **Education split:** Separate instruments for high-ed (`EDUCD` ≥ 65) and low-ed (`EDUCD` ≤ 64) immigrants.
- **Additional maps** across decades or for actual immigration change.

Grading Rubric

This rubric applies to **both paths**. Each category is assessed on a Baseline / Strong / Exceptional scale. Baseline corresponds to a satisfactory submission; Strong to good work; Exceptional to excellent.

Core (100%)

Category	Weight	Performance Standards
Data collection	25%	**Baseline:** Data is downloaded from the appropriate source and present in your submission. **Strong:** Correct samples and variables selected for all five decades; extract choices documented in the write-up. **Exceptional:** Data collection process described clearly enough for a classmate to replicate from scratch; data source limitations noted.
Data cleaning & wrangling	25%	**Baseline:** Code attempts the key operations—geography crosswalk merge, origin classification, sample restrictions, and collapse—for at least two decades. **Strong:** All five decades processed correctly; collapse produces the right unit of observation (CZ $\times$ origin, then CZ); shift-share instruments properly constructed and normalized for all decades. **Exceptional:** Multi-decade processing is automated (loops or functions rather than copy-paste); intermediate checks (observation counts, summary statistics) included; edge cases handled cleanly.
Estimation & visualization	25%	**Baseline:** A first-stage regression and a map of the instrument are produced. **Strong:** First-stage and IV wage regressions correctly specified with year FE and clustered SEs; map and scatterplots paneled by decade with clear labels. OLS vs. IV comparison interpreted in the write-up. **Exceptional:** Coefficient plot used to present regression results; interpretation demonstrates understanding of the instrument's economic logic and direction of OLS bias. Education split or additional specifications attempted.
Reproducibility & write-up	25%	**Baseline:** Code files and a write-up (any format) are submitted. **Strong:** Code runs end-to-end without errors; a log file documenting the full run is submitted. Write-up is organized, concise, and includes properly formatted figures and tables. **Exceptional:** Code includes a header listing dependencies, uses relative paths throughout, and is well-commented. Write-up reads like a professional short research note.

Extra Credit (up to +30%)

The following technical skills are **not required** for a good grade. You can earn full marks on the core rubric without them. However, they are valuable professional skills, and demonstrating them earns extra credit.

Skill	Bonus	What earns it
IPUMS API	+10%	Data collection automated via a Python script using `ipumspy`. Script is included and reproducible.
GitHub	+10%	Project lives in a GitHub repository with a meaningful commit history, sensible folder structure, a `README.md`, and a `.gitignore`.
LaTeX	+10%	Write-up compiled in LaTeX (or R Markdown/Quarto with PDF output). Source file included alongside the compiled PDF.

Appendix A: Shift-Share Methodology

The Immigration Shift-Share Instrument

Kim, Merritt, and Peri (2024) study how local exposure to global shocks—including immigration—affects the creation of "new work" (newly emerging job types) across U.S. commuting zones.[1] Because local immigration is endogenous to local labor-market conditions (immigrants may be drawn to booming areas), they construct a shift-share (Bartik) instrumental variable that predicts local immigration using only historical settlement patterns and national-level inflows.

Intuition. Immigrants from a given origin country tend to settle where earlier immigrants from the same country already live—driven by established ethnic networks, not current local demand. By combining each commuting zone's historical composition of immigrant origins with subsequent national immigration growth from those origins, the instrument predicts local immigration inflows that are driven by national trends rather than local economic conditions.

Construction. The instrument has two ingredients:

1. **Shares (from the 1970 Census).** For each commuting zone c and origin group o, compute:

$$s_{c,o,1970} \;=\; \frac{\text{Immigrants}_{c,o,1970}}{\text{Immigrants}^{\text{national}}_{o,1970}}$$

   This is the fraction of origin o's total U.S. immigrant population that was living in commuting zone c in 1970.

2. **Shifts (national growth by origin).** For each origin group o and decade t, compute the national change in immigrant population:

$$\Delta M_{o,t} \;=\; \text{Immigrants}^{\text{national}}_{o,t} \;-\; \text{Immigrants}^{\text{national}}_{o,t-1}$$

The shift-share instrument for commuting zone c in decade t is then:

$$\widetilde{B}_{c,t} \;=\; \frac{1}{\text{pop}_{c,1970}} \sum_{o} s_{c,o,1970} \;\times\; \Delta M_{o,t}$$

where $\text{pop}_{c,1970}$ is the total population of commuting zone c in 1970 (used for normalization).

Identification logic. The instrument is valid if the 1970 settlement shares are uncorrelated with decade-t local labor-demand shocks (conditional on controls). This is plausible because the shares are determined decades before the outcome period, and immigrant location choices in 1970 reflect historical network formation rather than current economic conditions. Following the framework of Goldsmith-Pinkham, Sorkin, and Swift (2020), identification comes from the exogeneity of the historical shares.

[1] Kim, Gueyon, Cassandra Merritt, and Giovanni Peri. "Measuring and Predicting 'New Work' in the United States: The Role of Local Factors and Global Shocks." NBER Working Paper No. 32526, May 2024.

Data. The instrument is constructed from U.S. Decennial Census microdata obtained through IPUMS USA, covering the 1970, 1980, 1990, 2000, and 2010 Censuses. Individual-level data are mapped to commuting zones (1990 definitions) using crosswalks from Dorn (2009), and birthplace codes are mapped to approximately 80 origin-country groups using a custom crosswalk.

Appendix B: Checkpoint Statistics

Use these to verify you are on the right track at each stage of the pipeline. Small deviations are acceptable if you selected a different IPUMS subsample; large deviations (wrong order of magnitude, wrong sign, wrong number of observations) indicate a coding error.

After Step 1

Check the number of observations in each saved file. These are CZ × origin pairs—if your count is much lower, you may have an incorrect merge key or a failed `joinby`. If your count is much higher, you may not have collapsed fully.

Decade	Observations (CZ × origin pairs)
1970	15,983
1980	28,258
1990	30,796
2000	35,685
2010	22,173

The count grows from 1970 to 2000 because later decades have more immigrant-origin diversity per CZ. The 2010 count is lower because the ACS is a smaller sample than the 5% Census.

Check a population total. After collapsing, `sum pop` across all CZ × origin pairs in your 1980 file should be approximately 163,550,000. If your total is orders of magnitude larger, you may have used `(sum)` instead of `(rawsum)` with probability weights.

After Step 2

Check the dimensions. You should have 82 origin groups after dropping US-born (`kmp_ctry == 78`) and other/unspecified (`kmp_ctry == 82`). If you have 84, you forgot to drop. If you have far fewer, the origin merge may have failed.

Check a national total. Mexico (`kmp_ctry` labeled MEX) should have a 1970 national immigrant total of approximately 721,000 and a 1980 total of approximately 1,829,000. The 1980 shift (change from 1970 to 1980) should be approximately +1,108,000—a large positive inflow. If your Mexican shift is negative, your decade subtraction may be reversed.

After Step 3

Check the dimensions. You should have 3,610 observations = 722 CZs × 5 decades. If you have fewer CZs, check whether your `joinby` dropped unmatched geographic units. If you have only 4 decades, you may have dropped 1970 too early—keep it here (you need the 1970 levels for computing changes in Step 5).

Check a CZ total. The largest CZ by 1970 population should have `total_pop` around 6.4 million (this is the Los Angeles commuting zone). Its 1990 immigrant count should be approximately 3.4 million.

After Step 4

Check the dimensions. You should have 2,888 observations = 722 CZs × 4 decades. There is no 1970 row because the shift-share instrument requires a *change* in national immigration, and 1970 is the base year.

Check the instrument's sign and scale. The mean shift-share instrument should be positive in every decade (immigration to the US grew overall). Values should be small fractions (the instrument is normalized by 1970 population). If your values are in the thousands or millions, you forgot to normalize by `pop1970`.

Decade	imm_shift (mean)	imm_shift (sd)
1980	0.011	0.028
1990	0.022	0.050
2000	0.046	0.106
2010	0.035	0.072

After Step 5

Check the dimensions. Still 2,888 observations. The merge should be `1:1` on `czone` and `year` with no unmatched observations from either file.

Check the full summary by decade. Use `tabstat imm_shift imm_actch wage wage_ch, by(year)` to verify:

Decade	imm_shift (mean)	imm_actch (mean)	wage (mean)	wage_ch (mean)
1980	0.011	0.012	5.88	2.75
1990	0.022	0.016	9.35	3.47
2000	0.046	0.048	13.63	4.28
2010	0.035	0.044	17.54	3.91

The shift-share instrument (predicted) and actual immigration change should have similar means in each decade—this is the first-stage relationship you will estimate in Step 6. If they have opposite signs, check whether your shift computation (Step 2) or your actual change computation (Step 5) has the decade subtraction reversed.